MAAT v1.2.1 Observable Growth Signature Proxy

Projection-Modulated $f\sigma_8$, Emergent Robustness, and a Boundary-Limited Diagnostic Scan

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Abstract

This paper tests a minimal observable-signature proxy for the MAAT v1.2.1 framework in the growth observable $f\sigma_8(z)$. Building on the projection observable developed in the preceding cosmology papers, we modulate a Planck-normalised ΛCDM growth baseline by a bounded projection template $\hat{\mathcal{C}}_{\rm proj}(z)$,

$$f\sigma_{8,\rm MAAT}(z) = f\sigma_{8,\Lambda\rm CDM}(z)\left[1+\epsilon\,\hat{\mathcal{C}}_{\rm proj}(z)\right].$$

The construction is extended with the MAAT v1.2.1 robustness closure,

$$R_{\rm resp} = (HBV)^{1/3}, \qquad R_{\rm rob} = \min\Big(R_{\rm resp}, (HBSV)^{1/4}\Big),$$

and with CCI diagnostics that avoid double counting of the implicit respect sector.

For a compact 13-point $f\sigma_8$ comparison set, a scan over $\epsilon \in [-0.01, 0.01]$ gives a weak improvement from $\chi^2_{\Lambda\rm CDM} = 12.4373$ to $\chi^2_{\rm MAAT} = 12.3772$. The best value lies at the lower scan boundary, $\epsilon = -0.01$, so this is not interpreted as a parameter measurement or evidence for modified growth. Rather, the result is a controlled diagnostic: a sub-percent MAAT projection signature can be inserted into a growth observable without producing large residuals or violating the v1.2.1 closure logic.

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1 Purpose and Status

Paper 38 tested the v1.2.1 robustness closure inside a linear-growth benchmark and found sub-percent growth-level deviations. The present paper asks a narrower follow-up question:

> *Can the MAAT projection observable be used as a small, explicit signature template for $f\sigma_8(z)$ while preserving the v1.2.1 closure structure?*

The status of the calculation is deliberately limited. It is not a Boltzmann-code result, not a full perturbation-theory treatment, and not a precision cosmological likelihood. It is a reproducible observable-signature proxy designed to test whether the projection layer can be coupled to a measured growth observable in a stable and interpretable way.

2 Reference Growth Baseline

We use a flat Planck-normalised reference cosmology

$$H_0 = 67.4, \qquad \Omega_{m0} = 0.315, \qquad \Omega_{\Lambda 0} = 0.685, \qquad \sigma_{8,0} = 0.811. \tag{1}$$

The reference expansion function is

$$E_{\Lambda\mathrm{CDM}}(z) = \sqrt{\Omega_{m0}(1+z)^3 + \Omega_{\Lambda 0}}, \tag{2}$$

and the matter fraction is

$$\Omega_m(z) = \frac{\Omega_{m0}(1+z)^3}{E^2_{\Lambda\mathrm{CDM}}(z)}. \tag{3}$$

For the compact diagnostic comparison, the growth factor is approximated by a Carroll–Press–Turner style expression, normalised to $D(0) = 1$. The reference growth observable is then

$$f_{\Lambda\mathrm{CDM}}\sigma_8(z) = \Omega_m(z)^\gamma\, D(z)\, \sigma_{8,0}, \qquad \gamma = 0.55. \tag{4}$$

This is sufficient for a controlled proxy calculation, but it is not a replacement for a full perturbation solver.

3 Projection Signature Template

The MAAT projection template follows the projection logic used in Papers 33 and 34. First, define an expansion/activity contribution

$$B_{\mathrm{exp}}(z) = E_{\Lambda\mathrm{CDM}}(z)(1+z), \tag{5}$$

and a bounded latent-depth suppression

$$L(z) = L_{\min} + \frac{1 - L_{\min}}{1 + \exp[s(z - z_c)]}, \qquad L_{\min} = 0.20, \quad s = 3.5, \quad z_c = 1.1. \tag{6}$$

The effective depth term is

$$D_{\mathrm{acc}}(z) = D(z)L(z), \tag{7}$$

and the bounded projection activation is

$$Q(z) = \tanh\left[\frac{B_{\mathrm{exp}}(z)}{\max B_{\mathrm{exp}}}\right]. \tag{8}$$

The projection observable is

$$\mathcal{C}_{\rm proj}^{\rm raw}(z) = \frac{B_{\rm exp}(z)Q(z)}{1 + D_{\rm acc}(z) + \varepsilon}. \tag{9}$$

The normalised projection observable used below is

$$\mathcal{C}_{\rm proj}(z) = \frac{\mathcal{C}_{\rm proj}^{\rm raw}(z)}{\mathcal{C}_{\rm proj}^{\rm raw}(0)}. \tag{10}$$

The transition proxy is estimated from the maximal interior curvature of the associated imbalance profile $R_{\rm proj}$:

$$z_{\rm tr} = \arg\max_z \left| \frac{{\rm d}^2 R_{\rm proj}}{{\rm d}z^2} \right|. \tag{11}$$

The reproduced value is

$$z_{\rm tr} = 0.6508. \tag{12}$$

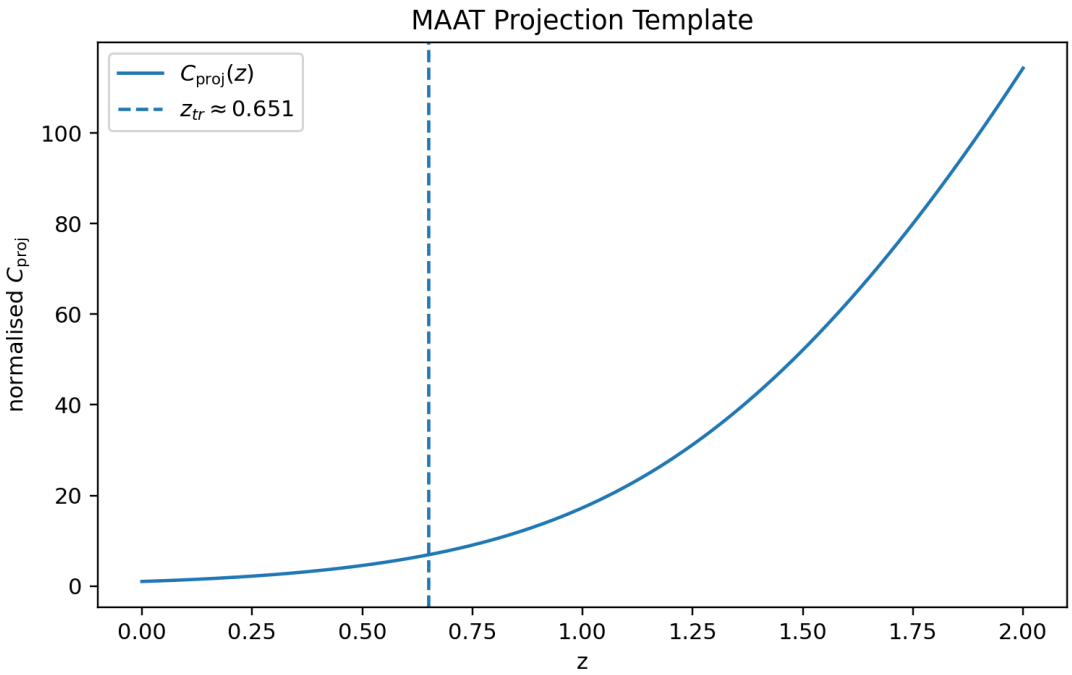

Figure 1: MAAT projection template $\mathcal{C}_{\rm proj}(z)$ and curvature-based transition proxy. The transition estimate is a diagnostic marker, not a cosmological parameter fit.

4 Observable Signature Proxy

The observable proxy is

$$\boxed{f_{\rm MAAT}\sigma_8(z) = f_{\Lambda\rm CDM}\sigma_8(z)\left[1 + \epsilon\,\hat{\mathcal{C}_{\rm proj}}(z)\right].} \tag{13}$$

Here $\hat{\mathcal{C}_{\rm proj}}$ is a bounded, standardised version of $\log(1 + \mathcal{C}_{\rm proj})$:

$$\hat{\mathcal{C}_{\rm proj}} = \tanh\left(\frac{\log(1 + \mathcal{C}_{\rm proj}) - \langle \log(1 + \mathcal{C}_{\rm proj})\rangle}{{\rm std}[\log(1 + \mathcal{C}_{\rm proj})] + \varepsilon}\right). \tag{14}$$

The scan range is deliberately small:

$$\epsilon \in [-0.01, 0.01]. \tag{15}$$

Thus the proxy tests percent-level signature compatibility rather than a large deformation of the growth sector.

The goodness-of-fit diagnostic is

$$\chi^2(\epsilon) = \sum_i \left[\frac{f_{\mathrm{MAAT}}\sigma_8(z_i;\epsilon) - f_{\mathrm{obs}}\sigma_8(z_i)}{\sigma_i} \right]^2 . \tag{16}$$

The scan gives

$$\epsilon_{\mathrm{best}} = -0.0100, \qquad \chi^2_{\Lambda\mathrm{CDM}} = 12.4373, \qquad \chi^2_{\mathrm{MAAT}} = 12.3772, \qquad \Delta\chi^2 = -0.0601. \tag{17}$$

Because the optimum lies at the lower boundary of the chosen scan interval, it should be read as a weak directional diagnostic rather than a stable best-fit parameter.

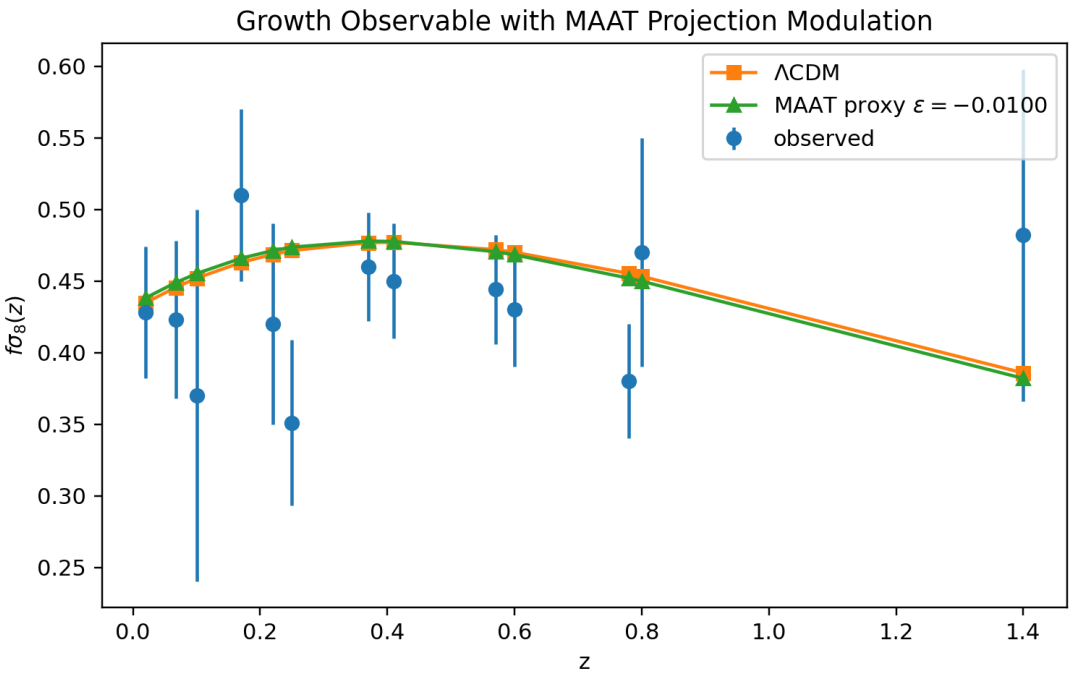

Figure 2: Compact $f\sigma_8$ comparison. The MAAT proxy introduces a small projection-modulated deviation from the ΛCDM reference curve. The points are used for diagnostic comparison only.

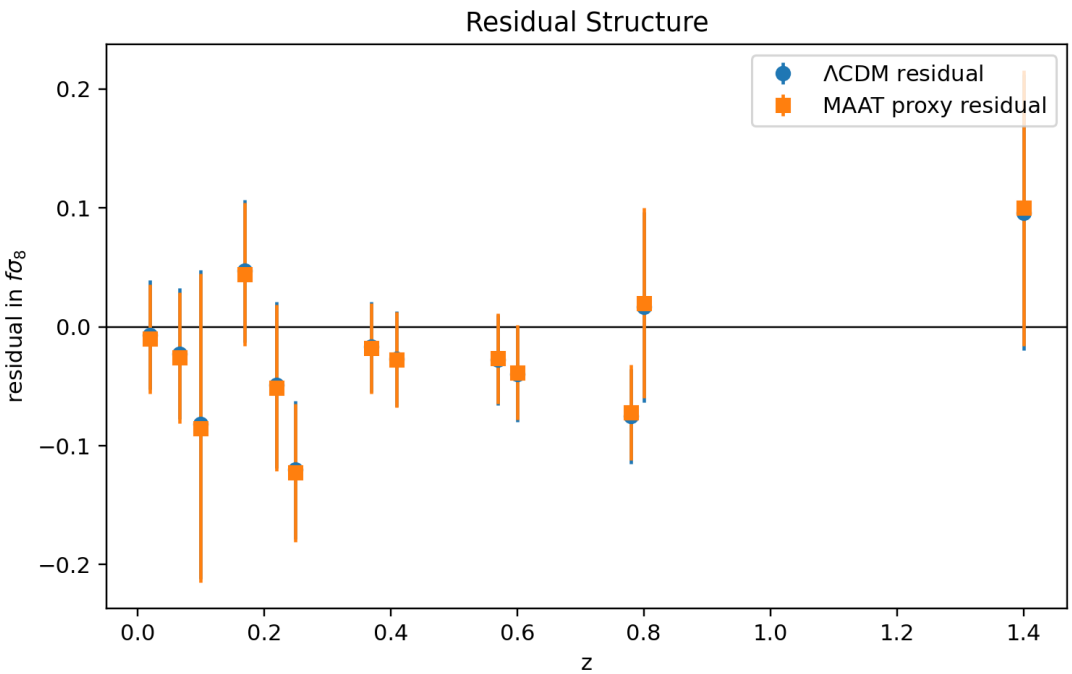

Figure 3: Residuals relative to the compact growth comparison set. The MAAT proxy changes the residual pattern only mildly.

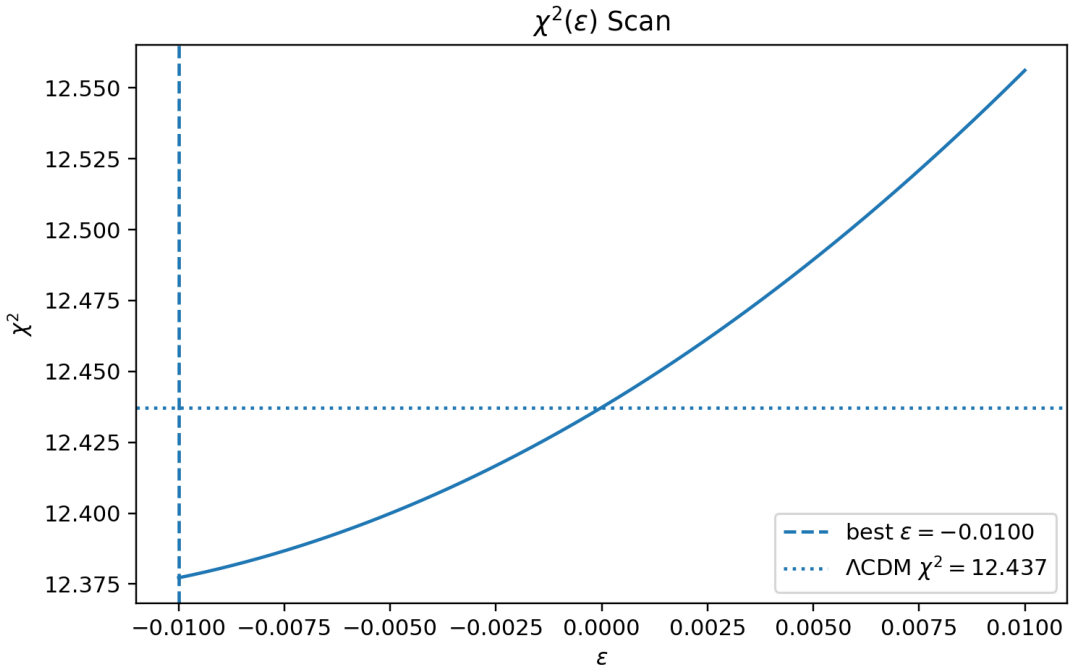

Figure 4: $\chi^2(\epsilon)$ scan. The minimum occurs at the lower boundary of the imposed scan window, which prevents interpreting ϵ as a measured parameter.

The maximum and mean fractional deviations are

$$\max\left|\frac{\Delta(f\sigma_8)}{f\sigma_8}\right| = 0.9891\%, \qquad \left\langle\left|\frac{\Delta(f\sigma_8)}{f\sigma_8}\right|\right\rangle = 0.5725\%. \tag{18}$$

5 v1.2.1 Robustness Closure

The v1.2.1 convention treats (H, B, S, V) as the primary support sectors and constructs robustness as an emergent closure rather than a fifth independent primary sector. For non-negative diagnostic defects,

$$\Gamma_a = \frac{1}{1 + d_a}, \tag{19}$$

the Paper 39 proxy uses:

- H: projection smoothness/coherence support,
- B: data-balance support from fractional $f\sigma_8$ residuals,
- S: projection-activity support,
- V: connectedness support from bounded projection imbalance.

These are diagnostic proxy supports, not fundamental field measurements.

The closure is

$$\boxed{R_{\text{resp}} = (HBV)^{1/3}, \qquad R_{\text{rob}} = \min\Big(R_{\text{resp}}, (HBSV)^{1/4}\Big).} \tag{20}$$

The stability convention is

$$\text{Stability} = R_{\text{rob}}. \tag{21}$$

The two CCI variants are

$$\text{CCI}_{\text{min}} = \frac{S}{H + B + V + \varepsilon}, \qquad \text{CCI}_{\text{diag}} = \frac{S}{H + B + V + R_{\text{rob}} + \varepsilon}. \tag{22}$$

The implicit respect support R_{resp} is not included separately in the CCI denominator. This avoids double counting, because R_{resp} is already a function of the primary support sectors.

Quantity	Mean value on growth comparison points
R_{resp}	0.7247
R_{rob}	0.6673
CCI_{min}	0.2662
CCI_{diag}	0.2036

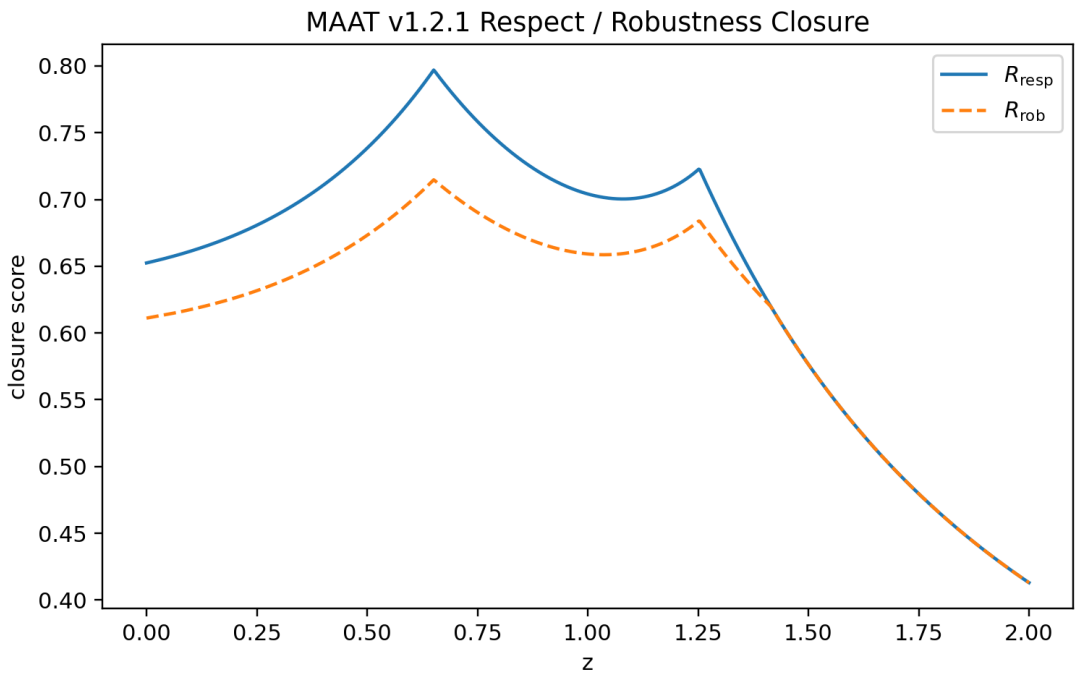

Figure 5: v1.2.1 respect and robustness closure on the projection grid. Robustness is bounded by both implicit respect and the full four-sector geometric mean.

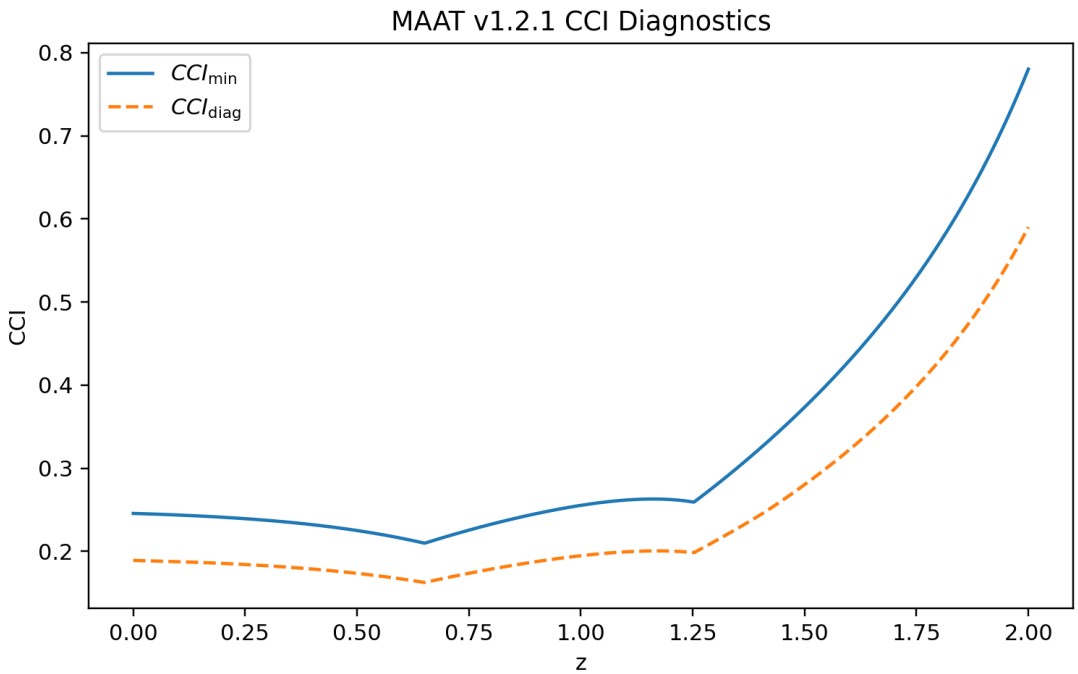

Figure 6: Minimal and robustness-weighted CCI diagnostics. The diagnostic CCI is lower because R_{rob} contributes to the stabilising denominator.

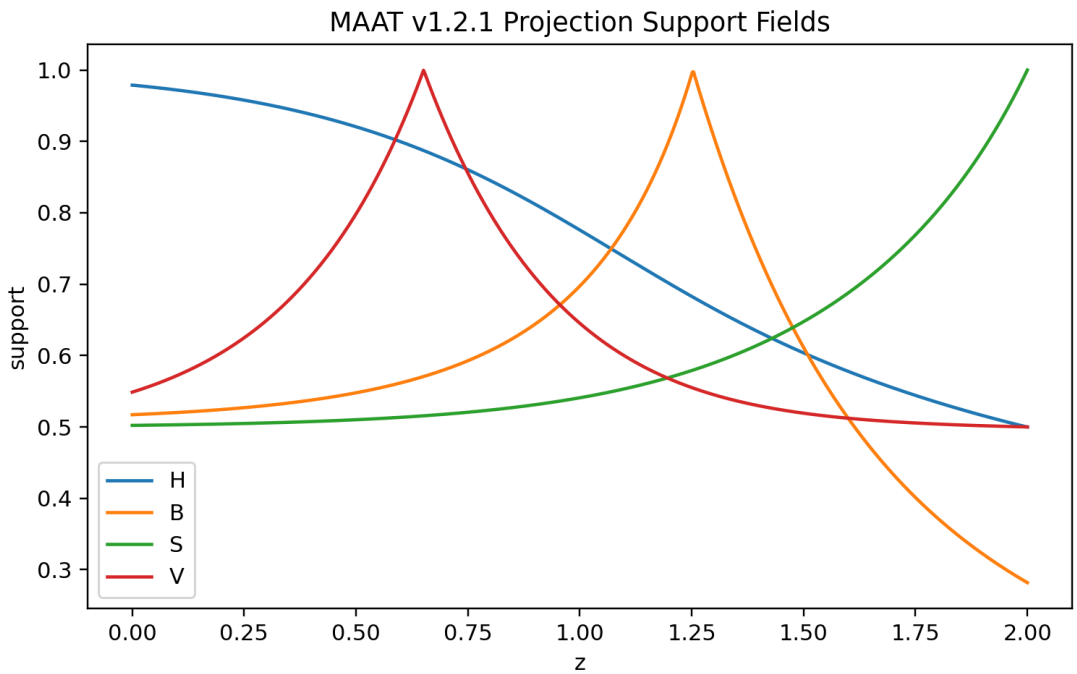

Figure 7: Projection support fields (H, B, S, V) on the redshift grid. The supports should be read as operational diagnostics for this benchmark.

6 Interpretation

The calculation has three main implications. First, the MAAT projection layer can be inserted into an observable growth quantity at the sub-percent level without creating large residuals. Second, the v1.2.1 closure remains operational in a data-facing proxy: the implicit respect support and the emergent robustness bound are computable from the primary support sectors. Third, the boundary-limited value of ϵ shows that the present calculation is not a parameter inference result.

The key scientific reading is therefore conservative:

> *Paper 39 demonstrates observable-signature compatibility, not observational preference.*

7 Limitations

- The growth factor uses an analytic approximation rather than a Boltzmann-code perturbation calculation.
- The compact $f\sigma_8$ comparison set is used diagnostically and does not include a full covariance matrix.
- The best ϵ value lies at the boundary of the imposed scan range, so it is not a robust parameter estimate.
- The support sectors (H, B, S, V) are operational proxy supports for this benchmark, not fundamental field measurements.
- The result does not establish evidence for modified gravity, altered dark energy, or a detection of MAAT cosmology.

8 Reproducibility

The numerical pipeline is contained in the public repository:

https://github.com/Chris4081/structural-selection-principle

The experiment folder is:

`experiments/maat_paper39_observable_growth_signature/`

Run:

```
python3 maat_paper39_observable_signature_v121.py
```

The script generates:

- `outputs_paper39/paper39_summary.json`,
- `outputs_paper39/paper39_growth_signature_results.csv`,
- `outputs_paper39/paper39_epsilon_chi2_scan.csv`,
- `outputs_paper39/paper39_projection_template_v121.csv`,
- `outputs_paper39/fig1_projection_template.png`,
- `outputs_paper39/fig2_growth_comparison.png`,
- `outputs_paper39/fig3_residuals.png`,
- `outputs_paper39/fig4_chi2_scan.png`,
- `outputs_paper39/fig5_v121_robustness_closure.png`,
- `outputs_paper39/fig6_v121_cci.png`,
- `outputs_paper39/fig7_v121_support_fields.png`.

9 Data Attribution and License Notes

The Planck-normalised reference parameters are external scientific results and should be cited to the Planck Collaboration when reused or discussed. The compact $f\sigma_8$ comparison points are literature-derived growth measurements used here only as a diagnostic comparison set. They are not redistributed as an authoritative survey catalogue, and no endorsement by the original survey collaborations or data authors is implied.

The CSV, JSON, and PNG files generated by this repository are derived analysis artifacts for reproducibility of the MAAT proxy calculation. The repository code is released under the repository license.

10 Conclusion

Paper 39 provides a small but important data-facing check of the MAAT projection program. A bounded projection observable can modulate $f\sigma_8(z)$ at the sub-percent level, remain compatible with a compact growth comparison set, and preserve the v1.2.1 robustness closure without double counting the implicit respect sector.

The result is not a detection. It is a reproducible observable-signature benchmark that prepares the next stage: replacing the proxy modulation by a full perturbation-level model and testing it with proper multi-probe likelihoods.

References

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[7] C. Krieg, "The Maturation of MAAT: From Projection Observables (v0.11) to Structural Selection (v1.0) to Emergent Robustness Closure (v1.2.1)," preprint (2026).

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